

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	(a)		Converting ly to m i.e. $200 \times 9.46 \times 10^{15} = 1.892 \times 10^{18}$ [m] (1) Calculating θ in radian i.e. $\frac{1.5 \times 10^{11}}{1.892 \times 10^{18}} = 7.93 \times 10^{-8}$ (1) (can be done by using tan or trig) $\theta = 1.22 \frac{\lambda}{d}$ used (whatever the subject) i.e. 7.6 m for visible light or 7.625×10^{-8} seen (1) Final calculation leading to suitable conclusion (1) Award full marks for a calculation of 1.45×10^{11} seen and compared with 1 AU	1	1 1		4	3	
	(b)	(i)	Substitution into period equation e.g. even $50 = 2\pi \sqrt{\frac{d^3}{G \times 3 \times 2 \times 10^{30}}}$ (1) Final rearrangement and correct substitution seen or 2.93×10^{12} [m] (1)	1	1		2	1	
		(ii)	Substitution into C of M equation (1) can be implied $\frac{1}{3} \times 2.93 \times 10^{12} = 0.977 \times 10^{12}$ or equivalent seen (1)	1	1		2	1	
		(iii)	Use of Stefan's law (1) Use of area proportional to R^2 (1) $\frac{R_A^2 T_A^4}{R_B^2 T_B^4} = \frac{R_A^2 1.7^4}{R_B^2 4.3^4} = \frac{25}{1/18}$ (1) Answer = $\frac{R_{\text{Sirius A}}}{R_{\text{Sirius B}}} = 136$ or 140 (1)	1	1 1 1		4	3	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)	Constant intensity when both stars seen or gradual drop or partial eclipse or regular time intervals (1) One star in front of the other, light is “blocked” (1) Different dips depending on which star is “blocked” (1)			3	3		
		(ii)	{Intensity / power / luminosity / brightness} increases with T (1) The same area is blocked regardless of which star is doing the blocking or $I = \frac{P}{A} = \sigma T^4$ (1)			2	2		
	(d)		Black hole is not visible [and star orbiting is]			1	1		
	(e)		Reflection method (1) accept the first method of the 3 Black bodies do not reflect OR black bodies absorb all em radiation incident (1) Award one mark only for existence of absorption lines in the spectrum			2	2		
			Question 7 total	4	8	8	20	8	0